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Polytechnic University of the Philippines Santa Rosa Campus

Bachelor of Science in Information Technology

COMP 006 Data Structures and Algorithms

NOVA GEN-AI CHATBOT

Final Project Documentation

Azir, Mac Justin

Benjamin, Edmigail D.

Cabral, Matt Clyde

Magyaya, Kyle R.

Peñada, Rhen Iverson

Villanueva, John Carlo T.

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Introduction

The Nova Generative Artificial Intelligence (Gen-AI) chatbot is an advanced artificial intelligence (AI) system designed to enhance human-computer interactions through intelligent responses and automation.

Built using cutting-edge machine learning models, Nova Gen-AI leverages application programming interfaces (APIs) to integrate with powerful AI engines like Gemini, thus enabling seamless natural language understanding.

Whether used for answering queries, generating content, or assisting with complex tasks, Nova Gen-AI provides an intuitive and efficient experience with its ability to learn and adapt—representing a step forward in the evolution of AI-driven technology.

Thus, the Nova Gen-AI chatbot is built to simplify tasks, improve efficiency, and deliver an efficient user experience.

Functions

The Nova Gen-AI chatbot has a wide range of functions due to being a large multi-modal model. Those functions are as follows but are not limited to :

- a. **Natural Language Processing (NLP)**, which includes;
 - Text generation and translation
 - Question answering and summarization
 - Sentiment analysis and emotion recognition



- Named entity recognition and relationship extraction
- b. **Computer Vision**, which includes:
- Image classification, object detection, and segmentation
 - Facial recognition and emotion detection
 - Image editing and enhancement
- c. **Audio Processing**, which includes:
- Speech recognition and generation
 - Speaker identification and language identification
 - Audio classification and transcription
- d. **Machine Learning**, which includes:
- Supervised and unsupervised learning
 - Model training and evaluation
 - Predictive analytics and forecasting
- e. **Data Analysis**, which includes:
- Data cleaning and preprocessing
 - Exploratory data analysis and visualization
 - Statistical modeling and hypothesis testing

Nova's other capabilities include but are not limited to: *(f)* knowledge base and information retrieval, *(g)* contextual understanding and reasoning, *(h)* problem-solving and decision-making, and *(i)* code generation and debugging.



However, Nova still has its limitations as one cannot fully replicate the wonder that is the human brain through artificial means. The Nova Gen-AI Chatbot's limitations may include; (1) the lack of common sense knowledge and the ability to make intuitive judgments, (2) the lack of creativity in generating original ideas, (3) the lack of access to real-time information— as its knowledge cutoff was from as far as back in April 2023, and (4) the exhibition of biases in its responses due to the training data it was fed on.

Beneficiaries

Being a Gen-AI chatbot, Nova does not have a specified sample of users like social media platforms or online forums might. It was designed to interact with a wide range of users, from individuals seeking information or assistance to researchers and developers exploring the capabilities of artificial intelligence (AI).

The intended users of the Nova Gen-AI Chatbot are the following, but not limited to:

a. Individuals seeking information or assistance

These users may have questions about a wide range of topics; such as current events, weather, sports, or travel. They may also be looking for help with tasks such as setting alarms, making appointments, or finding local businesses.

b. Developers and researchers

These users are interested in exploring the capabilities of AI and developing new applications for chatbot technology. They may ask questions about the underlying



algorithms and techniques used by chatbots, or they may provide feedback and suggestions for improvements.

c. Students and educators

These users use chatbots for educational purposes, such as getting help with homework, practicing language skills, or learning about new subjects.

d. Business people and organizations

These users use chatbots to provide customer support, answer frequently asked questions, or automate tasks such as scheduling appointments or processing orders.

Overall, the Nova Gen-AI Chatbot is designed to be a versatile and accessible tool for a wide range of users, striving to provide helpful and informative responses to its beneficiaries.

Source code for the chatbot

Below is the source code that powers the Nova Gen-AI Chatbot. The chatbot is written in the Python programming language and utilizes API to import Google's Gemini AI as its core language model. The script then sends the user queries to Gemini AI API, which processes the input and returns relevant responses.

By leveraging the API, the program can access advanced NLP capabilities without requiring complex and extensive local computation. Furthermore, the API keeps the application up to date with the latest advancements in AI technology. This integration allows Nova AI to deliver accurate, context-aware responses effectively.



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```
import google.generativeai as genai

genai.configure(api_key="AIzaSyDhDu-Vn9nTQndGaRDtVjrh0Vll8CPedto")

print("HI, I'M NOVA!\n")

def chat_with_gemini (user_input):

    model = genai.GenerativeModel ("gemini-pro" )

    response = model.generate_content(user_input)

    return response.text

while True:

    user_input = input("You: ")

    if user_input.lower() in ["quit", "exit", "bye"]:

        print("Chatbot Nova: Goodbye!")

        break

    print("Chatbot Nova:", chat_with_gemini(user_input))
```

Disclaimer: To run the Nova Gen-AI, you will need to install a gen-AI extension in your IDE's terminal by typing “*pip install google-generativeai*” or “*pip install google.generativeai*”.

Output

Below is the output of the Nova Gen-AI Chatbot.



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HI, I'M NOVA!

You: Hello

Chatbot Nova: Hello there! How can I assist you today?

You: Can you give me some study tips?

Chatbot Nova: ****Creating an Effective Study Environment****

- * ****Establish a dedicated study space:**** Minimize distractions by finding a quiet, comfortable, and well-lit area.
- * ****Set a schedule and stick to it:**** Plan out specific study times and avoid procrastination.
- * ****Take breaks:**** Step away from your work periodically to clear your head and recharge.

****Active Learning Strategies****

- * ****Spaced repetition:**** Review material at increasing intervals to strengthen memory.
- * ****Active recall:**** Try to remember information without looking at notes, then check your accuracy.
- * ****Elaboration:**** Connect new concepts to existing knowledge and create mental frameworks.
- * ****Retrieval practice:**** Regularly test yourself on what you've learned to identify areas that need improvement.

****Organization and Note-Taking****

- * ****Organize your materials:**** Use a binder, folders, or electronic notes to keep track of your work.
- * ****Take effective notes:**** Use a note-taking system that works for you, such as outlining, mind mapping, or Cornell notes.
- * ****Summarize and condense:**** Create concise summaries of key points to aid in review.

****Study Techniques****

- * ****Pomodoro Technique:**** Work for 25-minute intervals, followed by 5-minute breaks.
- * ****Interleaving:**** Study different subjects or topics in alternation to improve retention.
- * ****SQ3R:**** Preview, question, read, recite, and review to actively engage with material.
- * ****Concept mapping:**** Create visual diagrams that connect concepts and show relationships.

****Additional Tips****

- * ****Get enough sleep:**** Rest is essential for memory consolidation.
- * ****Maintain a healthy diet:**** Nourish your brain with nutrient-rich foods.
- * ****Form study groups:**** Collaborate with peers to discuss concepts and reinforce learning.
- * ****Use technology wisely:**** Utilize flashcards, note-taking apps, and online resources to enhance studying.
- * ****Seek help when needed:**** Don't hesitate to reach out to a tutor, teacher, or classmate if you encounter difficulties.
- * ****Reward yourself:**** Celebrate your progress and stay motivated with small rewards.



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Reward yourself. Celebrate your progress and stay motivated with small rewards.

You: Create an inspirational quotes.

Chatbot Nova: "Embrace the challenges of life, for they are the seeds of growth and the foundation of resilience."

You: What about a beautiful poem.

Chatbot Nova: **Ode to Serenity**

In verdant meadows, where whispers dance,
And nature's symphony fills the trance,
There lies a haven, a tranquil retreat,
Where serenity weaves its threads so sweet.

The willows weep with emerald grace,
Their branches sway in nature's embrace,
Like gentle maidens, they softly bow,
Whispering secrets to the wind's gentle flow.

The wildflowers bloom in vibrant hues,
A vibrant tapestry, enchanting views,
Their petals unfold, a fragrant dream,
Inviting weary souls to pause and glean.

The babbling brook, a silver stream,
Meanders through the meadow, a liquid gleam,
Its soothing sound, a lullaby's caress,
Washes over hearts, bringing sweet blessedness.

The air is filled with tranquility's might,
A balm for troubled souls, a beacon of light,
It whispers gently, like a lover's plea,
"Come, weary heart, and find respite with me."

In this sacred space, time seems to pause,
As worries vanish, and the soul withdraws,
Into a realm of peace, where cares dissolve,
And serenity's embrace, all hearts involve.

So let us seek this haven, far from strife,
Where beauty blooms and whispers new life,
In nature's sanctuary, we shall find,
The solace and tranquility we long to bind.



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You: What is a Data Structures and Algorithms about?

Chatbot Nova: ****Data Structures****

- * Ways to organize and store data in memory efficiently
- * Examples: arrays, linked lists, queues, stacks, trees, graphs

****Algorithms****

- * Step-by-step procedures for solving computational problems
- * Examples: sorting, searching, optimization, graph traversal

****Relationship between Data Structures and Algorithms****

- * Data structures provide the underlying storage mechanism for data.
- * Algorithms operate on data stored in data structures and perform computations.

****Key Concepts in Data Structures and Algorithms****

- * ****Complexity analysis:**** Measuring the efficiency of algorithms (time and space requirements)
- * ****Recursion:**** Solving problems by breaking them down into smaller instances
- * ****Abstraction:**** Hiding implementation details to focus on functionality
- * ****Robustness:**** Ensuring algorithms handle invalid inputs and edge cases
- * ****Optimization:**** Improving the performance of algorithms by reducing complexity

****Benefits of Studying Data Structures and Algorithms****

- * Develops problem-solving skills
- * Enhances coding efficiency
- * Improves understanding of computer science fundamentals
- * Prepares for competitive programming and interviews
- * Applications in various fields, including:
 - * Software development
 - * Data science
 - * Machine learning
 - * Bioinformatics

You: Thank you!

Chatbot Nova: You're welcome! Is there anything else I can help you with today?

You: None, Goodbye!

Chatbot Nova: It was nice talking to you too. Goodbye!



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Conclusion:

Written in the Python programming language through the use of Google Gemini's API, the Nova Gen-AI is a chatbot designed to enhance human-to-computer interaction. Although Nova is not designed for a specific user, it is however beneficial to individuals who seek out information quickly, or to developers to learn how machine learning works. It is made from the combined knowledge of other AI engines which allows seamless machine learning.

However, Nova is incapable of creating an original idea. Its access to information has also been paused in April 2023 which means Nova is unaware of current events that happened past the date it was shut down. For an AI, it also has a sense of bias due to the data it was fed upon which hinders it from making intuitive judgments.